

VƯƠNG VIỆT HÙNG

+84-583-885-391 | vuongviethung156@gmail.com

 [vuong-viet-hung](#) |  [vuong-viet-hung](#) |  0009-0006-8041-7608

Hanoi, Vietnam - 11025, Vietnam

EDUCATION

- Soongsil University** March 2024 - March 2026
Master's Degree in Electronic Engineering Seoul, South Korea
 - GPA: 4.24/4.50
- Hanoi University of Science and Technology** August 2019 - September 2023
Bachelor's Degree in Control Engineering and Automation Hanoi, Vietnam
 - GPA: 3.34/4.00

EXPERIENCE

- VinDynamics** February 2026 – September 2026
SLAM & Navigation Engineer Hanoi, Vietnam

Developed and deployed SLAM and height mapping systems for humanoid robotic platforms, with a focus on sensor integration, system evaluation, and optimization.
- ANDA Lab, Soongsil University** March 2024 – February 2026
Research Assistant Seoul, South Korea

Conducted research in LiDAR-based 3D computer vision for autonomous driving and contributed to peer-reviewed publications, researcher mentoring, and technical seminars.
- Rikai Mind** August 2022 – January 2023
AI Engineer Intern Hanoi, Vietnam

Developed and deployed a computer vision system for object volume estimation from monocular images and camera-object distance.
- Mandevices Laboratory, Hanoi University of Science and Technology** August 2020 – August 2023
Research Assistant Hanoi, Vietnam

Conducted research in deep-learning-based bearing fault diagnosis and contributed to data collection, publications, and researcher mentoring.

PUBLICATIONS

- Domain Adaptive LiDAR Panoptic Segmentation via Pseudo-Supervised Instance Learning** May 2026
First author – IEEE Transactions on Intelligent Transportation Systems
DOI: [XX.XXXX/XXXXX](#)
 - Proposed PILLA, a domain adaptation framework that generates instance-level pseudo-labels via unsupervised clustering and applies a noise-aware loss for robust pseudo-supervision.
 - Achieved state-of-the-art performance on nuScenes, outperforming the prior domain adaptation baseline by 10.97 percentage points in panoptic quality (PQ).
- Instance Embedding as Queries for DETR-based LiDAR Panoptic Segmentation** October 2024
Co-first author – IEEE Transactions on Intelligent Vehicles
DOI: [XX.XXXX/XXXXX](#)
 - Proposed a DETR-based architecture that uses learned instance embeddings as object queries for LiDAR panoptic segmentation.
 - Achieved state-of-the-art performance on nuScenes, outperforming the previous best method by 1.20 percentage points in panoptic quality (PQ).
- Generalized Simulation-Based Domain Adaptation Approach for Intelligent Bearing Fault Diagnosis** 2024
Co-author – Arabian Journal for Science and Engineering, Vol. 49, Issue 12, pp. 16941–16957
DOI: [10.1007/s13369-024-09282-1](#)

- Developed a simulation-based domain adaptation approach that generates synthetic vibration signals from bearing geometry to address data scarcity in intelligent bearing fault diagnosis.
- Achieved 74–98% fault classification accuracy across different bearing types using a CNN trained on synthesized vibration signals.

• **ConvMamba: A Data-Efficient Neural Network for Bearing Fault Diagnosis**

July 2024

Co-author – 2024 7th International Conference on Green Technology and Sustainable Development (GTSD)

DOI: [10.1109/GTSD62346.2024.10674880](https://doi.org/10.1109/GTSD62346.2024.10674880)

- Developed ConvMamba, a data-efficient architecture combining convolutional networks and Vision Mamba for bearing fault diagnosis.
- Improved classification accuracy by 27 percentage points over the predecessor Hybrid-CNN-MLP model on the CWRU bearing fault dataset.

PROJECTS

• **Visual SLAM Profiling and Latency Optimization**

August 2026

VinDynamics

- Profiled the OKVIS2 visual SLAM pipeline and identified landmark matching and local-window optimization as major bottlenecks, taking approximately 20 ms and 40 ms respectively.
- Evaluated lower-latency front-end pose publishing, observing only 1.16 cm RMSE in translational APE relative to back-end poses.
- Redesigned frame scheduling to prioritize the latest available frame and drop stale frames, reducing average front-end latency from 124.00 ms to 47.00 ms and back-end latency from 164.30 ms to 87.64 ms.

• **Multi-Sensor Local Height Mapping for Humanoid Robot**

July 2026 – August 2026

VinDynamics

- Extended an existing LiDAR-based GPU height mapper to integrate Livox LiDAR and Intel RealSense depth data, implementing sensor synchronization, rate limiting, pose lookup, and robot self-filtering.
- Implemented and benchmarked pose extrapolation to compensate for SLAM latency; linear extrapolation reduced translational APE RMSE from 2.148 cm to 0.445 cm (79.3%) for a 30 Hz depth camera.
- Deployed and validated the multi-sensor height mapping system on an in-house humanoid robot with calibrated LiDAR and depth camera data.

• **Stereo Matching and 3D Reconstruction**

March 2023 – August 2023

Graduation thesis, Hanoi University of Science and Technology

- Implemented and compared block matching and deep learning approaches for dense stereo matching on SceneFlow (EPE: 77.32%, D1: 79.04%).
- Reconstructed 3D scenes from estimated disparity maps via triangulation and deployed the pipeline on a Jetson Nano-based wheeled robot.

• **Object Volume Estimation from Images**

August 2022 – December 2022

Rikai Mind

- Developed an object volume estimation model from monocular images and camera–object distance, achieving an R^2 score of 84.78%.
- Extracted visual embeddings with CNNs, localized objects using keypoint regression, and fused visual features with camera–object distance for volume regression; deployed the model using FastAPI.

SKILLS

- **Programming languages:** Python, C/C++, MATLAB
- **Frameworks:** PyTorch, TensorFlow, OpenCV, ROS2, Kalibr (for sensor calibration)
- **Hardware platforms:** Jetson Orin Nano, Jetson Nano, Raspberry Pi
- **Sensors:** Intel RealSense camera, ZED X camera, Livox LiDAR
- **Development and deployment tools:** Git, GitHub, GitLab, Docker, FastAPI, Milvus
- **Others:** LaTeX

LANGUAGES

- **English:** IELTS 7.5
- **Vietnamese:** Native speaker